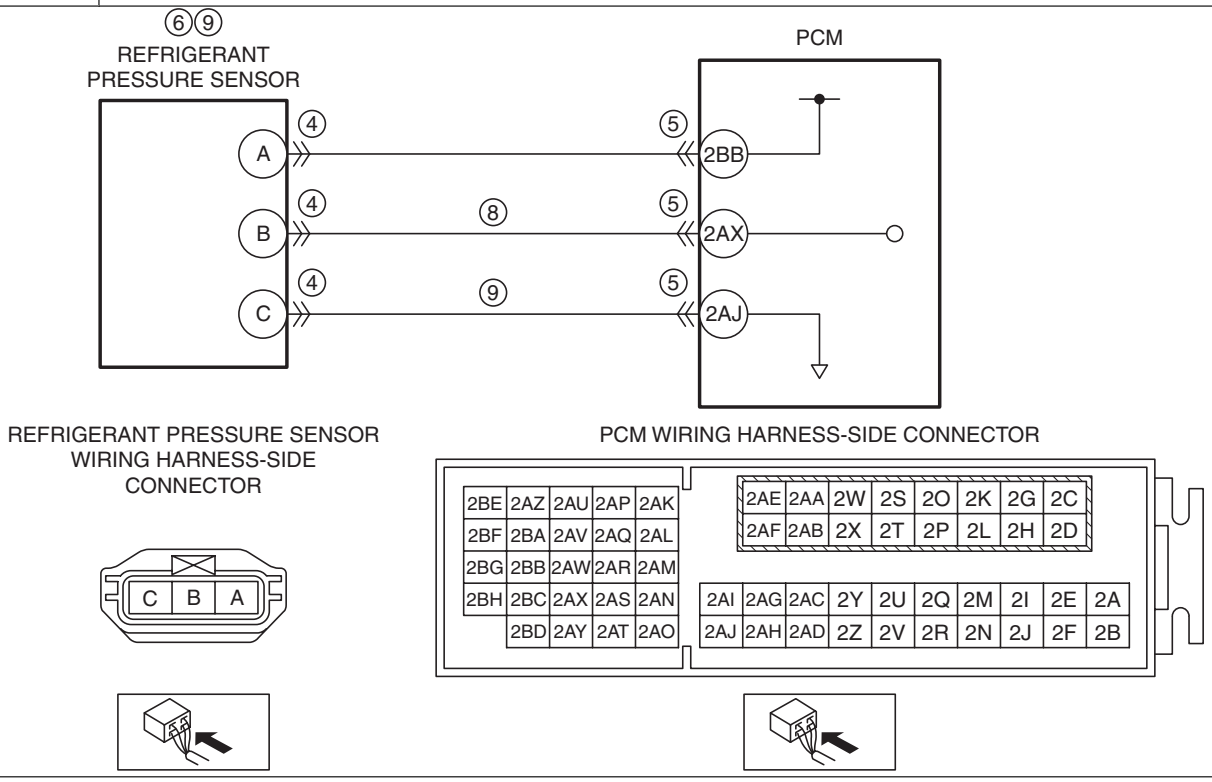
DTC P0533:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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DTC P0533:00	Refrigerant pressure sensor circuit high input
DETECTION CONDITION	• The PCM monitors input voltage from the refrigerant pressure sensor. If the input voltage at the PCM terminal 2AX is **above 4.9 V** for **5 s**, the PCM determines that the refrigerant pressure sensor circuit has a malfunction. **Diagnostic support note** • This is a continuous monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA is not available. • Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Refrigerant pressure sensor connector or terminals malfunction • PCM connector or terminals malfunction • Refrigerant pressure sensor malfunction • Short to power supply in wiring harness between refrigerant pressure sensor terminal B and PCM terminal 2AX • Open circuit in wiring harness between the following terminals: — Refrigerant pressure sensor terminal B—PCM terminal 2AX — Refrigerant pressure sensor terminal C—PCM terminal 2AJ • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the next step.

STEP	INSPECTION		ACTION
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**DETERMINE IF REFRIGERANT PRESSURE SENSOR OR WIRING HARNESS MALFUNCTION** • Access the AC_PRES PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Verify the AC_PRES PID value. • Is the AC_PRES PID value **5 V** or **B+**?	Yes	Go to Step 7.
		No	Go to the next step.
4	**INSPECT REFRIGERANT PRESSURE SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the refrigerant pressure sensor connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
5	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 10.
		No	Go to the next step.
6	**INSPECT REFRIGERANT PRESSURE SENSOR** • Reconnect all disconnected connectors. • Inspect the refrigerant pressure sensor. (See REFRIGERANT PRESSURE SENSOR INSPECTION [MANUAL AIR CONDITIONER].) (See REFRIGERANT PRESSURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is there any malfunction?	Yes	Replace the refrigerant pressure sensor, then go to Step 10. (See REFRIGERANT PRESSURE SENSOR REMOVAL/ INSTALLATION [MANUAL AIR CONDITIONER].) (See REFRIGERANT PRESSURE SENSOR REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].)
		No	Go to Step 10.
7	**DETERMINE IF REFRIGERANT PRESSURE SENSOR SIGNAL CIRCUIT OR REFRIGERANT PRESSURE SENSOR GROUND CIRCUIT MALFUNCTION** • Switch the ignition off. • Disconnect the refrigerant pressure sensor connector. • Access the AC_PRES PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Verify the AC_PRES PID value. • Is the AC_PRES PID value **5 V** or **B+**?	Yes	Go to the next step.
		No	Go to Step 9.

STEP	INSPECTION		ACTION
8	**INSPECT REFRIGERANT PRESSURE SENSOR SIGNAL CIRCUIT FOR OPEN CIRCUIT** • Verify that the refrigerant pressure sensor connector is disconnected. • Switch the ignition off. • Disconnect the PCM connector. • Inspect for continuity between refrigerant pressure sensor terminal B (wiring harness-side) and PCM terminal 2AX (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between refrigerant pressure sensor terminal B and PCM terminal 2AX. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 10.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between refrigerant pressure sensor terminal B and PCM terminal 2AX. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.
9	**INSPECT REFRIGERANT PRESSURE SENSOR GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the refrigerant pressure sensor connector is disconnected. • Switch the ignition off. • Disconnect the PCM connector. • Inspect for continuity between refrigerant pressure sensor terminal C (wiring harness-side) and PCM terminal 2AJ (wiring harness-side). • Is there continuity?	Yes	Replace the refrigerant pressure sensor, then go to the next step. (See REFRIGERANT PRESSURE SENSOR REMOVAL/INSTALLATION [MANUAL AIR CONDITIONER].) (See REFRIGERANT PRESSURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
		No	Refer to the wiring diagram and verify whether or not there is a common connector between refrigerant pressure sensor terminal C and PCM terminal 2AJ. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
10	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
11	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.